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Editors
NAR Genomics and Bioinformatics

Dear Editors,

Please consider the manuscript entitled "Layered representation diagnostics for short metagenomic reads" for publication as a Standard Paper in *NAR Genomics and Bioinformatics*.

Downstream accuracy summarizes task performance but does not directly characterize which sequence attributes remain accessible after compression and perturbation. This manuscript introduces a paired representation-diagnostic framework and a 222-dimensional, training-free representation, CK4P-MSP, that combines reverse-complement canonical 4-mer composition with global nucleotide-property-coded summaries and multi-scale positional property pooling. Its main methodological contribution is a block-decomposable audit construction in which these information channels can be examined under one controlled perturbation protocol.

The evidence combines a seven-group K/P/MSP ablation, shared-template length and perturbation sweeps, matched counterfactual controls, dimension-matched high-k vectors, PCA/SVD reductions, historical physicochemical descriptors and high-dimensional positional-readout references. The analyses associate the global property block primarily with reduced perturbation drift, the multi-scale positional block with grouped local-change readout, and the canonical k-mer block with composition-linked retrieval. PseKNC and full-position encodings define lower-drift and higher-positional-readout boundaries, respectively. ART and public CAMI resources then provide simulator-contextualized stability, coarse fixed-head label retention and an anonymous-read composition-shift probe beyond the controlled whole-genome-sequence grid.

The manuscript fits the journal's scope through its focus on reproducible sequence-representation methodology, quantitative benchmarking and public metagenomic resources. Source code, processed source tables, figure inputs and a manuscript-to-script map are maintained in an MIT-licensed versioned repository at `https://github.com/FairYJmrx/ultrashort-dna-representation-diagnostics`. The `v1.2.0` source and output snapshot is archived at `https://doi.org/10.5281/zenodo.21882250`; subsequent archived versions are linked by the Zenodo concept DOI `https://doi.org/10.5281/zenodo.21792340`.

This manuscript is original, has not been published and is not under consideration by another journal. No preprint has been posted as of the date of this letter. The study used only public reference and benchmark resources and computationally simulated sequences; no human participants, patient-level data or identifiable private information were involved. The authors declare no competing interests and received no specific funding for this work.

Thank you for considering this manuscript.

Sincerely,

Ruixiang Mei
Corresponding author, on behalf of all authors